

Gamma fn

$n! = n(n-1)!$ for all x except non-positive

$$f(x) = x f(x-1)$$

$$f(x) = \int_0^{\infty} g(t, x) dt$$

$$\int_0^{\infty} g(t, x) dt = x \cdot \int_0^{\infty} g(t, x-1) dt$$

$$g(t, x) = x \cdot g(t, x-1)$$

$$g(t, x) = t^{x-1} e^{-t}$$

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

$$\Gamma(x+1) = x!$$

Gamma dist

$$f(x; \alpha, \lambda) = \frac{\lambda^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} = \int_0^x f(u; \alpha, \lambda) du = \frac{\gamma(x, \lambda)}{\Gamma(\alpha)}$$

$$\mu = \frac{\alpha}{\lambda} \quad \text{var} = \frac{\alpha}{\lambda^2}$$

* conjugate prior for poisson & exp dist

apps to Model +ve real valued random var
such as time between events in poisson process
duration of a task or process
size of RV

* Reg Models

used as response dist

↓ if

Model $\begin{cases} +ve \\ cont \end{cases}$

- Model reln betwn dep variable & ind variables

↓ follow gamma dist

* using GLM with gamma response dist $\rightarrow$ rel $\begin{cases} dep \\ indep \end{cases}$

* Survival Times (+ve, cont)

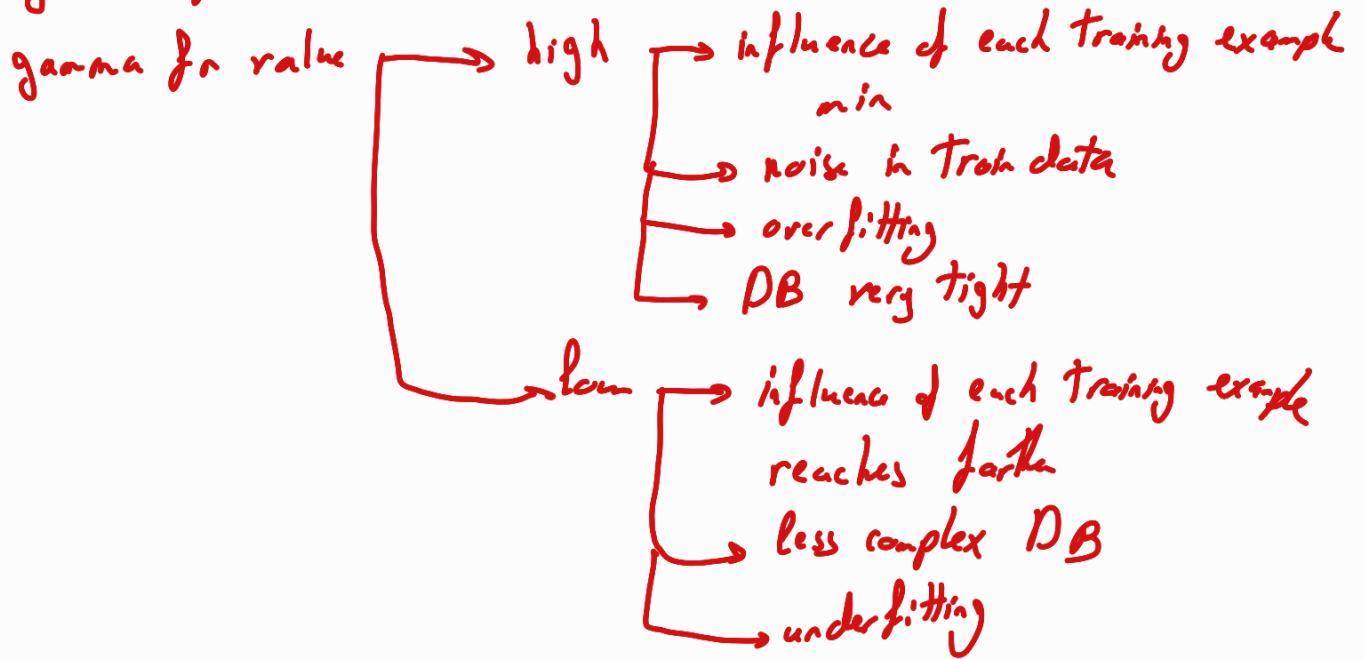
- time until patient experience certain event

- prop patient surviving beyond certain pt

* SVM

crucial hyperparameter (non-linear kernel)

gamma fn defines behaviour of decision boundary



Hessian Matrix

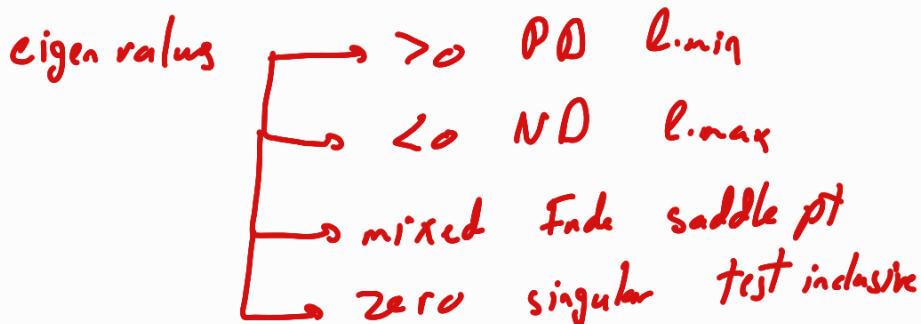
H: contain second partial der [curvature info]

in NN tell us How to take step

help in regularization (reach l.min)

not used in DL → computational power

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$



use its inverse to find critical pt

approximate errors

statistical diagnostics [fisher info matrix in max likelihood est]

CV det of Hessian (Delt) for feature detect